5 Risk Aversion and Investment Decisions

A Risk Aversion and Portfolio Allocation

B Portfolios, Risk Aversion, and Wealth

Risk Aversion and Portfolio Allocation

Let's now put our framework of decision-making under uncertainty to use.

Consider a risk-averse investor with vN-M expected utility who divides his or her initial wealth Y_0 into an amount a allocated to a risky asset – say, the stock market – and an amount $Y_0 - a$ allocated to a safe asset – say, a bank account or a government bond.

Risk Aversion and Portfolio Allocation

$Y_0 =$ initial wealth

$a =$ amount allocated to stocks

$\tilde{r} =$ random return on stocks

$r_f =$ risk-free return

$\tilde{Y}_1 =$ terminal wealth

$$\begin{aligned} \tilde{Y}_1 &= (1 + r_f)(Y_0 - a) + a(1 + \tilde{r}) \\ &= Y_0(1 + r_f) + a(\tilde{r} - r_f) \end{aligned}$$

Risk Aversion and Portfolio Allocation

The investor chooses a to maximize expected utility:

$$\max_a E[u(\tilde{Y}_1)] = \max_a E\{u[Y_0(1 + r_f) + a(\tilde{r} - r_f)]\}$$

If the investor is risk-averse, u is concave and the first-order condition for this unconstrained optimization problem, found by differentiating the objective function by the choice variable and equating to zero, is both a necessary and sufficient condition for the value a^* of a that solves the problem.

Risk Aversion and Portfolio Allocation

The investor's problem is

$$\max_a E\{u[Y_0(1+r_f)+a(\tilde{r}-r_f)]\}$$

The first-order condition is

$$E\{u'[Y_0(1+r_f)+a^*(\tilde{r}-r_f)](\tilde{r}-r_f)\}=0.$$

Note: we are allowing the investor to sell stocks short ($a^* < 0$) or to buy stocks on margin ($a^* > Y_0$) if he or she desires.

Risk Aversion and Portfolio Allocation

Theorem If the Bernoulli utility function u is increasing and concave, then

$$a^* > 0 \text{ if and only if } E(\tilde{r}) > r_f$$

$$a^* = 0 \text{ if and only if } E(\tilde{r}) = r_f$$

$$a^* < 0 \text{ if and only if } E(\tilde{r}) < r_f$$

Thus, a risk-averse investor will always allocate at least some funds to the stock market if the expected return on stocks exceeds the risk-free rate.

Risk Aversion and Portfolio Allocation

To prove the theorem, let

$$W(a) = E\{u'[Y_0(1 + r_f) + a(\tilde{r} - r_f)](\tilde{r} - r_f)\},$$

so that the investor's first-order condition can be written more compactly as

$$W(a^*) = 0.$$

Risk Aversion and Portfolio Allocation

Next, note that with

$$W(a) = E\{u'[Y_0(1 + r_f) + a(\tilde{r} - r_f)](\tilde{r} - r_f)\},$$

it follows that

$$W'(a) = E\{u''[Y_0(1 + r_f) + a(\tilde{r} - r_f)](\tilde{r} - r_f)^2\} < 0$$

since u is concave. This means that W is a decreasing function of a.

Risk Aversion and Portfolio Allocation

Finally, note that with

$$W(a) = E\{u'[Y_0(1+r_f) + a(\tilde{r} - r_f)](\tilde{r} - r_f)\},$$

$$\begin{aligned} W(0) &= E\{u'[Y_0(1+r_f) + 0(\tilde{r} - r_f)](\tilde{r} - r_f)\} \\ &= E\{u'[Y_0(1+r_f)](\tilde{r} - r_f)\} \\ &= u'[Y_0(1+r_f)]E(\tilde{r} - r_f) \\ &= u'[Y_0(1+r_f)][E(\tilde{r}) - r_f]. \end{aligned}$$

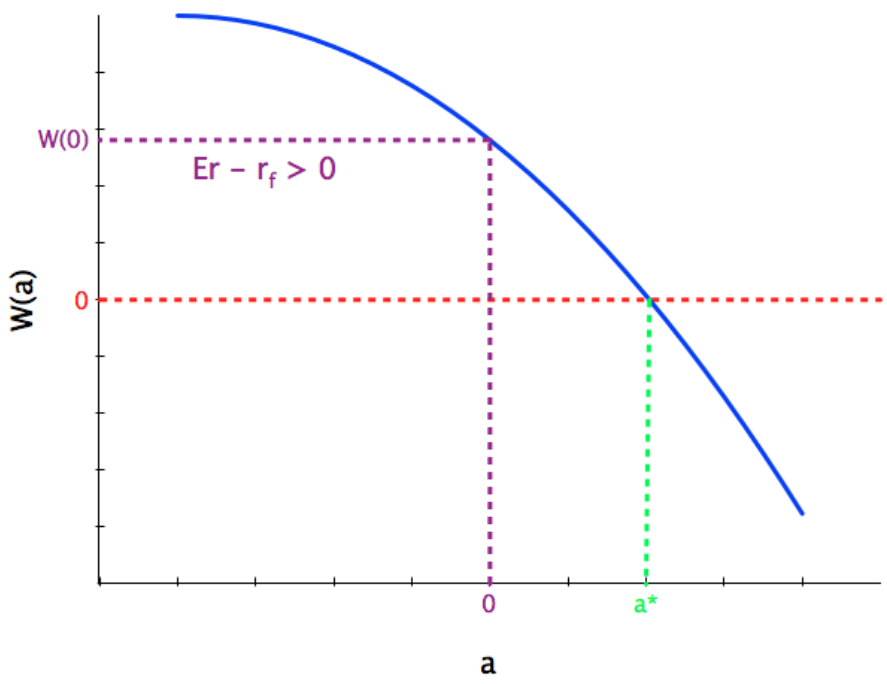
Since u is increasing, this means that $W(0)$ has the same sign as $E(\tilde{r}) - r_f$.

Risk Aversion and Portfolio Allocation

We now know that:

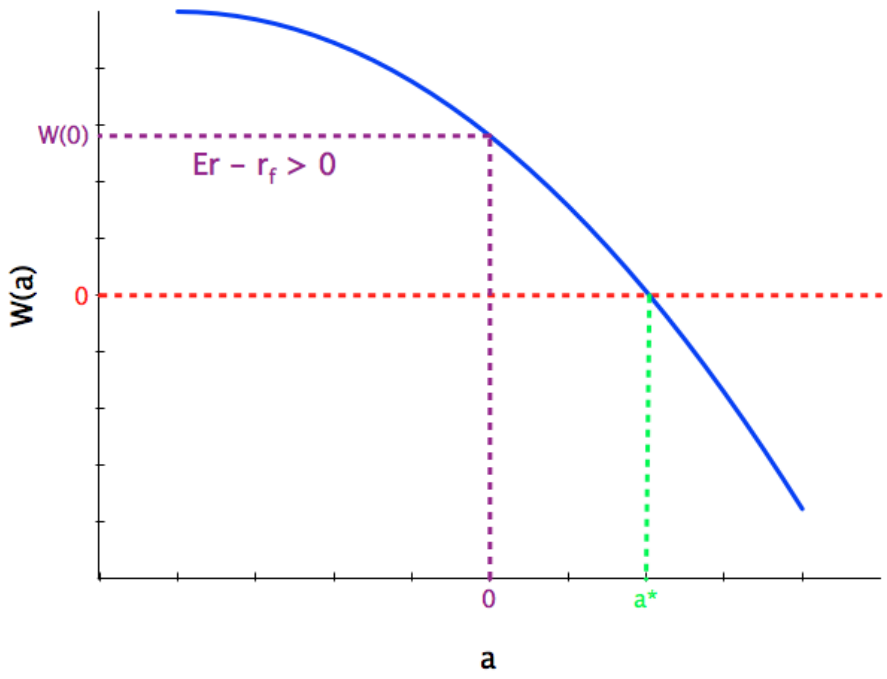
1. $W(a)$ is a decreasing function
2. $W(0)$ has the same sign as $E(\tilde{r}) - r_f$.
3. $W(a^*) = 0$

Risk Aversion and Portfolio Allocation



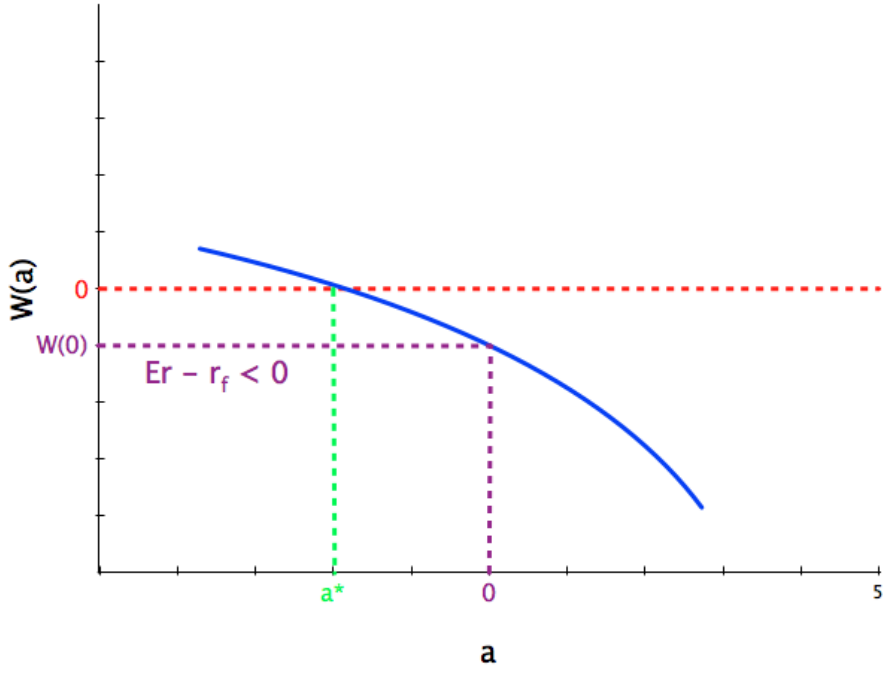
$E(\tilde{r}) - r_f > 0$ implies that $W(0) > 0$, and since W is decreasing, $W(a^*) = 0$ implies that $a^* > 0$.

Risk Aversion and Portfolio Allocation



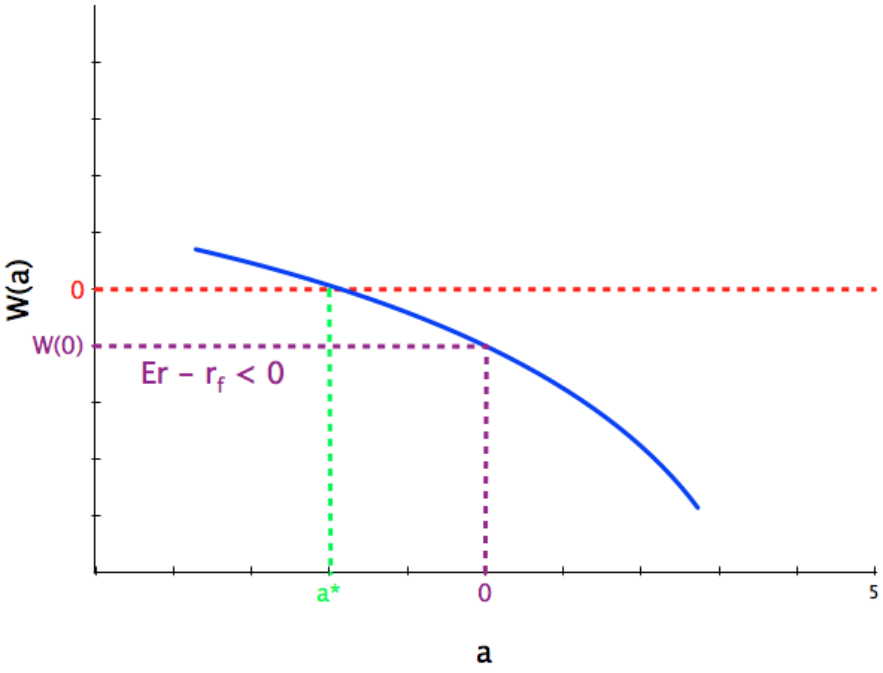
Since W is decreasing, $W(a^*) = 0$ and $a^* > 0$ imply that $W(0) > 0$. And since $W(0)$ has the same sign as $E(\tilde{r}) - r_f$, $E(\tilde{r}) - r_f > 0$.

Risk Aversion and Portfolio Allocation



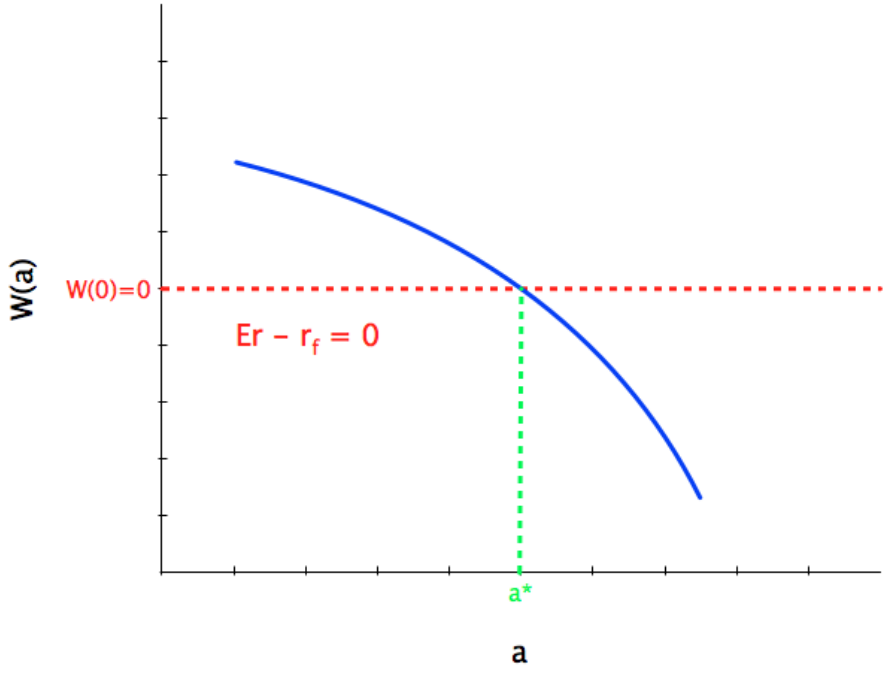
$E(\tilde{r}) - r_f < 0$ implies that $W(0) < 0$, and since W is decreasing, $W(a^*) = 0$ implies that $a^* < 0$.

Risk Aversion and Portfolio Allocation



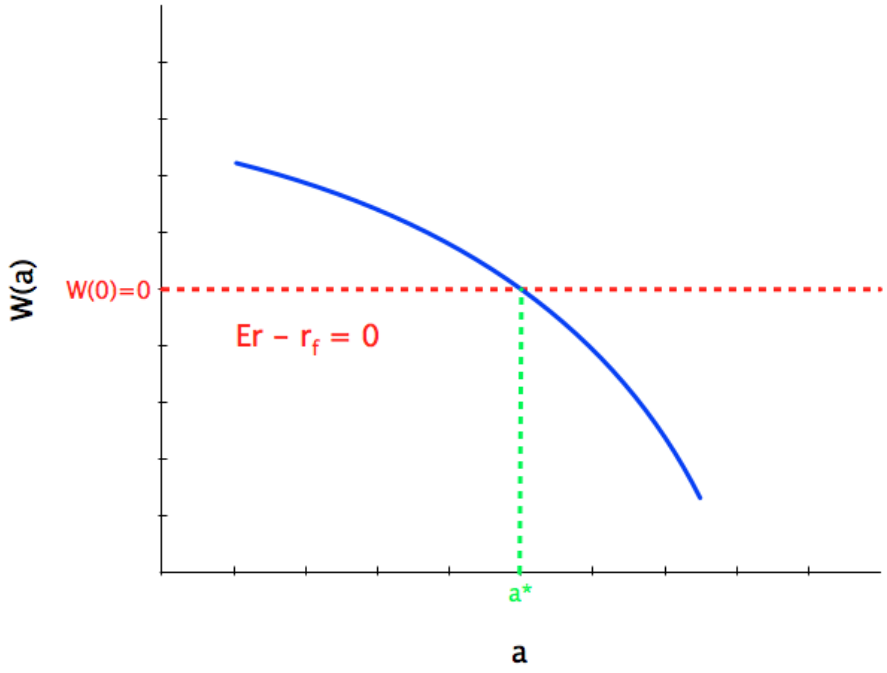
Since W is decreasing, $W(a^*) = 0$ and $a^* < 0$ imply that $W(0) < 0$. And since $W(0)$ has the same sign as $E(\tilde{r}) - r_f$, $E(\tilde{r}) - r_f < 0$.

Risk Aversion and Portfolio Allocation



$E(\tilde{r}) - r_f = 0$ implies that $W(0) = 0$, and since W is decreasing, $W(a^*) = 0$ implies that $a^* = 0$.

Risk Aversion and Portfolio Allocation



Since W is decreasing, $W(a^*) = 0$ and $a^* = 0$ imply that $W(0) = 0$. And since $W(0)$ has the same sign as $E(\tilde{r}) - r_f$, $E(\tilde{r}) - r_f = 0$.

Risk Aversion and Portfolio Allocation

Theorem If the Bernoulli utility function u is increasing and concave, then

$$a^* > 0 \text{ if and only if } E(\tilde{r}) > r_f$$

$$a^* = 0 \text{ if and only if } E(\tilde{r}) = r_f$$

$$a^* < 0 \text{ if and only if } E(\tilde{r}) < r_f$$

Thus, a risk-averse investor will always allocate at least some funds to the stock market if the expected return on stocks exceeds the risk-free rate.

Risk Aversion and Portfolio Allocation

Danthine and Donaldson (3rd ed., p.41) report that in the United States, 1889-2010, average real (inflation-adjusted) returns on stocks and risk-free bonds are

$$E(\tilde{r}) = 0.075 \text{ (7.5 percent per year)}$$
$$r_f = 0.011 \text{ (1.1 percent per year)}$$

The equity risk premium of $E(\tilde{r}) - r_f = 0.064$ (6.4 percent) is not only positive, it is huge. The implication of the theory is that all investors, even the most risk averse, should have some money invested in the stock market.

Risk Aversion and Portfolio Allocation

As an example, suppose $u(Y) = \ln(Y)$, as suggested by Daniel Bernoulli. Recall that for this utility function, $u'(Y) = 1/Y$. Then assume that stock returns can either be good or bad:

$$\tilde{r} = \begin{cases} r_G & \text{with probability } \pi \\ r_B & \text{with probability } 1 - \pi \end{cases}$$

where $r_G > r_f > r_B$ defines the "good" and "bad" states and

$$\pi r_G + (1 - \pi) r_B > r_f,$$

so that $E(\tilde{r}) > r_f$ and the investor will choose $a^* > 0$.

Risk Aversion and Portfolio Allocation

The problem

$$\max_a E\{u[Y_0(1+r_f)+a(\tilde{r}-r_f)]\}$$

specializes to

$$\begin{aligned}\max_a \quad & \pi \ln[Y_0(1+r_f)+a(r_G-r_f)] \\ & + (1-\pi)\ln[Y_0(1+r_f)+a(r_B-r_f)]\end{aligned}$$

Risk Aversion and Portfolio Allocation

The problem

$$\max_{a} \quad \pi \ln[Y_0(1+r_f) + a(r_G - r_f)] + (1-\pi)\ln[Y_0(1+r_f) + a(r_B - r_f)]$$

has first-order condition

$$\frac{\pi(r_G - r_f)}{Y_0(1+r_f) + a^*(r_G - r_f)} + \frac{(1-\pi)(r_B - r_f)}{Y_0(1+r_f) + a^*(r_B - r_f)} = 0.$$

Risk Aversion and Portfolio Allocation

$$\frac{\pi(r_G - r_f)}{Y_0(1+r_f) + a^*(r_G - r_f)} + \frac{(1-\pi)(r_B - r_f)}{Y_0(1+r_f) + a^*(r_B - r_f)} = 0$$

$$\begin{aligned} & \pi(r_G - r_f)[Y_0(1+r_f) + a^*(r_B - r_f)] \\ = \; & -(1-\pi)(r_B - r_f)[Y_0(1+r_f) + a^*(r_G - r_f)] \end{aligned}$$

$$\begin{aligned} & a^*(r_G - r_f)(r_B - r_f) \\ = \; & -Y_0(1+r_f)[\pi(r_G - r_f) + (1-\pi)(r_B - r_f)] \end{aligned}$$

Risk Aversion and Portfolio Allocation

$$\begin{aligned} & a^*(r_G - r_f)(r_B - r_f) \\ = \quad & -Y_0(1 + r_f)[\pi(r_G - r_f) + (1 - \pi)(r_B - r_f)] \end{aligned}$$

implies

$$\frac{a^*}{Y_0} = -\frac{(1 + r_f)[\pi(r_G - r_f) + (1 - \pi)(r_B - r_f)]}{(r_G - r_f)(r_B - r_f)},$$

which is positive, since $r_G > r_f > r_B$ and

$$E(\tilde{r}) - r_f = \pi(r_G - r_f) + (1 - \pi)(r_B - r_f) > 0.$$

Risk Aversion and Portfolio Allocation

$$\frac{a^*}{Y_0} = -\frac{(1+r_f)[\pi(r_G - r_f) + (1-\pi)(r_B - r_f)]}{(r_G - r_f)(r_B - r_f)},$$

In this case, a^*:

Rises proportionally with Y_0.

Increases as $E(\tilde{r}) - r_f$ rises.

Falls as r_G and r_B move father away from r_f, holding $E(\tilde{r})$ constant; that is, in response to a mean preserving spread.

Risk Aversion and Portfolio Allocation

$$\frac{a^*}{Y_0} = -\frac{(1+r_f)[\pi(r_G - r_f) + (1-\pi)(r_B - r_f)]}{(r_G - r_f)(r_B - r_f)},$$

r_f	r_G	r_B	π	$E(\tilde{r})$	a^*/Y_0
0.05	0.40	−0.20	0.50	0.10	0.60
0.05	0.30	−0.10	0.50	0.10	1.40
0.05	0.40	−0.20	0.75	0.25	2.40

The fraction of initial wealth allocated to stocks rises when stocks become less risky or pay higher expected returns.

Risk Aversion and Portfolio Allocation

Before moving on, return to the general problem

$$\max_a E\{u[Y_0(1+r_f)+a(\tilde{r}-r_f)]\}$$

but assume now that the investor is risk-neutral, with

$$u(Y) = \alpha Y + \beta,$$

and $\alpha > 0$, so that more wealth is preferred to less.

Risk Aversion and Portfolio Allocation

The risk-neutral investor solves

$$\begin{aligned} & \max_a E\{u[Y_0(1+r_f)+a(\tilde{r}-r_f)]\} \\ = & \max_a E\{\alpha[Y_0(1+r_f)+a(\tilde{r}-r_f)]+\beta\} \\ = & \max_a \alpha\{Y_0(1+r_f)+a[E(\tilde{r})-r_f]\}+\beta \end{aligned}$$

So long as $E(\tilde{r}) - r_f > 0$, the risk-neutral investor will choose a^* to be as large as possible, borrowing as much as he or she is allowed to in order to buy more stocks on margin.

Portfolios, Risk Aversion, and Wealth

The previous examples call out for a more detailed analysis of how optimal portfolio allocation decisions, summarized by the value of a^* that solves

$$\max_a E\{u[Y_0(1 + r_f) + a(\tilde{r} - r_f)]\}$$

are influenced by the investor's degree of risk aversion and his or her level of wealth.

Portfolios, Risk Aversion, and Wealth

The following result was proven by Kenneth Arrow in "The Theory of Risk Aversion," published in the 1971 volume *Essays in the Theory of Risk-Bearing* and reprinted in 1983 in volume 3 of the *Collected Papers of Kenneth J. Arrow* (Harvard University Press).

Theorem Consider two investors, $i = 1$ and $i = 2$, and suppose that for all wealth levels Y, $R_A^1(Y) > R_A^2(Y)$, where $R_A^i(Y)$ is investor i's coefficient of absolute risk aversion. Then $a_1^*(Y) < a_2^*(Y)$, where $a_i^*(Y)$ is amount allocated by investor i to stocks when he or she has initial wealth Y.

Portfolios, Risk Aversion, and Wealth

Recall that the coefficients of absolute and relative risk aversion are

$$R_A(Y) = -\frac{u''(Y)}{u'(Y)} \text{ and } R_R(Y) = -\frac{Yu''(Y)}{u'(Y)}.$$

Thus

$$R_A^1(Y) > R_A^2(Y) \text{ or } -\frac{u_1''(Y)}{u_1'(Y)} > -\frac{u_2''(Y)}{u_2'(Y)}$$

implies

$$-\frac{Yu_1''(Y)}{u_1'(Y)} > -\frac{Yu_2''(Y)}{u_2'(Y)} \text{ or } R_R^1(Y) > R_R^2(Y).$$

Portfolios, Risk Aversion, and Wealth

Arrow's result applies equally well to relative risk aversion:

Theorem Consider two investors, $i = 1$ and $i = 2$, and suppose that for all wealth levels Y, $R_R^1(Y) > R_R^2(Y)$, where $R_R^i(Y)$ is investor i's coefficient of relative risk aversion. Then $a_1^*(Y) < a_2^*(Y)$, where $a_i^*(Y)$ is amount allocated by investor i to stocks when he or she has initial wealth Y.

Portfolios, Risk Aversion, and Wealth

Let's test Arrow's proposition out, by generalizing our previous example with logarithmic utility to the case where

$$u(Y) = \frac{Y^{1-\gamma} - 1}{1 - \gamma},$$

with $\gamma > 0$. For this Bernoulli utility function, the coefficient of relative risk aversion is constant and equal to γ. The specific setting $\gamma = 1$ takes us back to the case with logarithmic utility.

Portfolios, Risk Aversion, and Wealth

Hence, in this extended example,

$$u(Y) = \frac{Y^{1-\gamma} - 1}{1-\gamma} \text{ implies } u'(Y) = Y^{-\gamma} = \frac{1}{Y^{\gamma}}.$$

and stock returns can either be good or bad

$$\tilde{r} = \begin{cases} r_G & \text{with probability } \pi \\ r_B & \text{with probability } 1-\pi \end{cases}$$

where $r_G > r_f > r_B$ defines the "good" and "bad" states and

$$\pi r_G + (1-\pi) r_B > r_f,$$

so that $E(\tilde{r}) > r_f$ and the investor will choose $a^* > 0$.

Portfolios, Risk Aversion, and Wealth

With CRRA (constant relative risk aversion) utility and two states for $\tilde{r}$, the problem

$$\max_a E\{u[Y_0(1+r_f)+a(\tilde{r}-r_f)]\}$$

specializes to

$$\begin{aligned}\max_a \quad & \pi\left\{\frac{[Y_0(1+r_f)+a(r_G-r_f)]^{1-\gamma}-1}{1-\gamma}\right\} \\ & +(1-\pi)\left\{\frac{[Y_0(1+r_f)+a(r_B-r_f)]^{1-\gamma}-1}{1-\gamma}\right\}\end{aligned}$$

Portfolios, Risk Aversion, and Wealth

The problem

$$\max_{a} \quad \pi \left\{ \frac{[Y_0(1+r_f) + a(r_G - r_f)]^{1-\gamma} - 1}{1-\gamma} \right\} + (1-\pi) \left\{ \frac{[Y_0(1+r_f) + a(r_B - r_f)]^{1-\gamma} - 1}{1-\gamma} \right\}$$

has first-order condition

$$\frac{\pi(r_G - r_f)}{[Y_0(1+r_f) + a^*(r_G - r_f)]^{\gamma}} + \frac{(1-\pi)(r_B - r_f)}{[Y_0(1+r_f) + a^*(r_B - r_f)]^{\gamma}} = 0.$$

Portfolios, Risk Aversion, and Wealth

$$\frac{\pi(r_G - r_f)}{[Y_0(1+r_f) + a^*(r_G - r_f)]^\gamma} + \frac{(1-\pi)(r_B - r_f)}{[Y_0(1+r_f) + a^*(r_B - r_f)]^\gamma} = 0$$

$$\begin{aligned} & \pi(r_G - r_f)[Y_0(1+r_f) + a^*(r_B - r_f)]^\gamma \\ = \; & (1-\pi)(r_f - r_B)[Y_0(1+r_f) + a^*(r_G - r_f)]^\gamma \end{aligned}$$

$$\begin{aligned} & [\pi(r_G - r_f)]^{1/\gamma}[Y_0(1+r_f) + a^*(r_B - r_f)] \\ = \; & [(1-\pi)(r_f - r_B)]^{1/\gamma}[Y_0(1+r_f) + a^*(r_G - r_f)] \end{aligned}$$

Portfolios, Risk Aversion, and Wealth

$$
\begin{aligned}
& [\pi(r_G - r_f)]^{1/\gamma}[Y_0(1+r_f) + a^*(r_B - r_f)] \\
= \; & [(1-\pi)(r_f - r_B)]^{1/\gamma}[Y_0(1+r_f) + a^*(r_G - r_f)]
\end{aligned}
$$

$$
\begin{aligned}
& Y_0(1+r_f)[\pi(r_G - r_f)]^{1/\gamma} + a^*(r_B - r_f)[\pi(r_G - r_f)]^{1/\gamma} \\
= \; & Y_0(1+r_f)[(1-\pi)(r_f - r_B)]^{1/\gamma} \\
& + a^*(r_G - r_f)[(1-\pi)(r_f - r_B)]^{1/\gamma}
\end{aligned}
$$

$$
\begin{aligned}
& Y_0(1+r_f)\{[\pi(r_G - r_f)]^{1/\gamma} - [(1-\pi)(r_f - r_B)]^{1/\gamma}]\} \\
= \; & a^*\{(r_G - r_f)[(1-\pi)(r_f - r_B)]^{1/\gamma} + (r_f - r_B)[\pi(r_G - r_f)]^{1/\gamma}\}
\end{aligned}
$$

Portfolios, Risk Aversion, and Wealth

$$
\begin{aligned}
& Y_0(1+r_f)\{[\pi(r_G-r_f)]^{1/\gamma}-[(1-\pi)(r_f-r_B)]^{1/\gamma}]\} \\
= \quad & a^*\{(r_G-r_f)[(1-\pi)(r_f-r_B)]^{1/\gamma}+(r_f-r_B)[\pi(r_G-r_f)]^{1/\gamma}\}
\end{aligned}
$$

implies

$$
\frac{a^*}{Y_0}=\frac{(1+r_f)\{[\pi(r_G-r_f)]^{1/\gamma}-[(1-\pi)(r_f-r_B)]^{1/\gamma}]\}}{(r_G-r_f)[(1-\pi)(r_f-r_B)]^{1/\gamma}+(r_f-r_B)[\pi(r_G-r_f)]^{1/\gamma}}
$$

Portfolios, Risk Aversion, and Wealth

$$\frac{a^*}{Y_0} = \frac{(1+r_f)\{[\pi(r_G - r_f)]^{1/\gamma} - [(1-\pi)(r_f - r_B)]^{1/\gamma}]\}}{(r_G - r_f)[(1-\pi)(r_f - r_B)]^{1/\gamma} + (r_f - r_B)[\pi(r_G - r_f)]^{1/\gamma}}$$

γ	r_f	r_G	r_B	π	$E(\tilde{r})$	a^*/Y_0
0.5	0.05	0.40	−0.20	0.50	0.10	1.20
1	0.05	0.40	−0.20	0.50	0.10	0.60
2	0.05	0.40	−0.20	0.50	0.10	0.30
3	0.05	0.40	−0.20	0.50	0.10	0.20
5	0.05	0.40	−0.20	0.50	0.10	0.12
10	0.05	0.40	−0.20	0.50	0.10	0.06

Portfolios, Risk Aversion, and Wealth

γ	r_f	r_G	r_B	π	$E(\tilde{r})$	a^*/Y_0
0.5	0.05	0.40	-0.20	0.50	0.10	1.20
1	0.05	0.40	-0.20	0.50	0.10	0.60
2	0.05	0.40	-0.20	0.50	0.10	0.30
3	0.05	0.40	-0.20	0.50	0.10	0.20
5	0.05	0.40	-0.20	0.50	0.10	0.12
10	0.05	0.40	-0.20	0.50	0.10	0.06

Consistent with Arrow's theorem, higher coefficients of relative risk aversion are associated with smaller values of a^*.

Portfolios, Risk Aversion, and Wealth

$$\frac{a^*}{Y_0} = \frac{(1+r_f)\{[\pi(r_G - r_f)]^{1/\gamma} - [(1-\pi)(r_f - r_B)]^{1/\gamma}]\}}{(r_G - r_f)[(1-\pi)(r_f - r_B)]^{1/\gamma} + (r_f - r_B)[\pi(r_G - r_f)]^{1/\gamma}}$$

Note that with constant relative risk aversion, a^* rises proportionally with wealth.

Two additional theorems, also proven by Arrow, tell us more about the relationship between a^* and wealth.

Portfolios, Risk Aversion, and Wealth

Theorem Let $a^*(Y_0)$ be the solution to

$$\max_a E\{u[Y_0(1+r_f)+a(\tilde{r}-r_f)]\}.$$

If $u(Y)$ is such that

(a) $R_A'(Y) < 0$ then $\frac{da^*(Y_0)}{dY_0} > 0$

(b) $R_A'(Y) = 0$ then $\frac{da^*(Y_0)}{dY_0} = 0$

(c) $R_A'(Y) > 0$ then $\frac{da^*(Y_0)}{dY_0} < 0$

Portfolios, Risk Aversion, and Wealth

Part (a)

$$R_A'(Y) < 0 \text{ then } \frac{da^*(Y_0)}{dY_0} > 0$$

describes the "normal" case where absolute risk aversion falls as wealth rises.

In this case, wealthier individuals allocate more wealth to stocks.

Portfolios, Risk Aversion, and Wealth

Part (b)

$$R_A'(Y) = 0 \text{ then } \frac{da^*(Y_0)}{dY_0} = 0$$

means that investors with constant absolute risk aversion

$$u(Y) = -\frac{1}{\nu} e^{-\nu Y}$$

allocate a constant amount of wealth to stocks.

This may seem surprising, but it reflects that fact that absolute risk aversion describes preferences over bets of a given size . . .

Portfolios, Risk Aversion, and Wealth

Part (b)

$$R_A'(Y) = 0 \text{ then } \frac{da^*(Y_0)}{dY_0} = 0$$

means that investors with constant absolute risk aversion

$$u(Y) = -\frac{1}{\nu} e^{-\nu Y}$$

allocate a constant amount of wealth to stocks.

...so a CARA investor finds a bet of the ideal size and sticks with it, even when income increases.

Portfolios, Risk Aversion, and Wealth

Part (c)

$$R_A'(Y) > 0 \text{ then } \frac{da^*(Y_0)}{dY_0} < 0$$

describes the case where absolute risk aversion rises as wealth rises.

The implication that wealthier individuals allocate less wealth to stocks makes this case seem less plausible.

Portfolios, Risk Aversion, and Wealth

Theorem Let $a^*(Y_0)$ be the solution to

$$\max_a E\{u[Y_0(1+r_f)+a(\tilde{r}-r_f)]\}.$$

If $u(Y)$ is such that

(a) $R_A'(Y) < 0$ then $\frac{da^*(Y_0)}{dY_0} > 0$

(b) $R_A'(Y) = 0$ then $\frac{da^*(Y_0)}{dY_0} = 0$

(c) $R_A'(Y) > 0$ then $\frac{da^*(Y_0)}{dY_0} < 0$

This result relates changes in absolute risk aversion to the absolute amount of wealth allocated to stocks.

Portfolios, Risk Aversion, and Wealth

Consistent with our results with CRRA utility, the next result relates changes in relative risk aversion to changes in the proportion of wealth allocated to stocks.

Define the elasticity of the function $a^*(Y_0)$ as

$$\eta = \frac{d \ln a^*(Y_0)}{d \ln Y_0} = \frac{Y_0}{a^*(Y_0)} \frac{da^*(Y_0)}{dY_0}$$

The elasticity measures the percentage change in a^* brought about by a percentage-point change in Y_0.

Portfolios, Risk Aversion, and Wealth

Theorem Let $a^*(Y_0)$ be the solution to

$$\max_a E\{u[Y_0(1+r_f)+a(\tilde{r}-r_f)]\}.$$

If $u(Y)$ is such that

(a) $R_R'(Y) < 0$ then $\eta > 1$

(b) $R_R'(Y) = 0$ then $\eta = 1$

(c) $R_R'(Y) > 0$ then $\eta < 1$

The theorem confirms what we know about CRRA utility: it implies that a^* rises proportionally with Y_0.

Portfolios, Risk Aversion, and Wealth

With CRRA utility:

$$\frac{a^*}{Y_0} = K$$

where

$$K = \frac{(1+r_f)\{[\pi(r_G - r_f)]^{1/\gamma} - [(1-\pi)(r_f - r_B)]^{1/\gamma}]\}}{(r_G - r_f)[(1-\pi)(r_f - r_B)]^{1/\gamma} + (r_f - r_B)[\pi(r_G - r_f)]^{1/\gamma}}.$$

Hence

$$\ln(a^*(Y_0)) = \ln(K) + \ln(Y_0)$$

and

$$\eta = \frac{d \ln a^*(Y_0)}{d \ln Y_0} = 1.$$

Portfolios, Risk Aversion, and Wealth

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But the theorem extends the results to the cases of decreasing and increasing relative risk aversion.